Chapter 3 — Data breaches

A data breach is unauthorized access to, disclosure of, or destruction of sensitive information

Learning objectives

- Define a data breach, name common causes
- Distinguish three organizational failure modes illustrated by Equifax, Yahoo
- Target

Understanding data breaches

- Excessive third-party access
- Organizations storing identity
- Statute-level notification clocks assume teams can detect and classify incidents quickly

Example 3.22 — Data breach severity range

- Example 3.22 — hands-on module
- Example 3.22 states the severity range
- Impact scales with sensitivity, volume, dwell time before detection
- Explore the chapter example module
- View files: `modules/chapter3/example22/`

Equifax — unpatched internet-facing systems

- Example 3.23 describes the 2017 Equifax incident
- The failure mode is delayed patching and weak monitoring on high-value identity stores
- Downstream costs appear in the next part

Example 3.23 — Equifax case study

- Example 3.23 — hands-on module
- The patching-discipline lesson
- Explore the chapter example module
- View files: `modules/chapter3/example23/`

Yahoo — delayed disclosure

- Example 3.24 covers Yahoo breaches affecting hundreds of millions of accounts
- The failure mode is late accountability to users and markets after compromise at consumer

Target — third-party vendor path

- Example 3.25 describes attackers using third-party vendor access to reach point-of-sale
- The failure mode is vendor risk and malware on retail networks

Takeaways

- Breach risk is organizational as much as cryptographic
- Three patterns recur, unpatched systems, delayed disclosure, and vendor-mediated entry
- Prevention controls mapped to each pattern come next

Next

- Complete the quiz for this part
- Continue to breach consequences and prevention